VOL. I · NO. 3 MONDAYS & THURSDAYS

THE LAUNCH REPORT

Rocket launches, reentries and the business of spaceflight, worldwide

MONDAY, OCTOBER 19, 2026 ALL TIMES EASTERN DAYLIGHT TIME (EDT)

THE WEEK IN LAUNCH

One launch since Thursday; H3 sends MMX toward Mars this afternoon

The only orbital launch anywhere since Thursday's issue was the classified USSF-481 mission, which a Falcon 9 lifted from Vandenberg at 8:01 p.m. EDT Thursday. That brings 2026 to 243 orbital launch attempts, 236 of them successful, according to GCAT. Bandwagon-5, set for this morning, slipped to Tuesday because of weather in the booster recovery zone, SpaceX posted.

Japan's H3 is due to launch the Martian Moons eXploration (MMX) probe at 3:41:03 p.m. EDT today, and the Japan Aerospace Exploration Agency (JAXA) has extended its reserve period to November 9. Eight more orbital launches are dated through Saturday: four Chinese, three by SpaceX and one by Rocket Lab. Rocket Lab pushed the first Neutron flight to 2027.

I · RECENT LAUNCHES & RETURNS

USSF-481 heads south from California

The classified Space Force mission was the 243rd orbital launch attempt of the year and the 119th by SpaceX.



A Falcon 9 lifts off from Space Launch Complex 4E at Vandenberg Space Force Base with NASA's SWOT satellite on Dec. 16, 2022. USSF-481 flew from the same pad on Thursday.

PHOTO: NASA/KEEGAN BARBER · PUBLIC DOMAIN

Falcon 9: USSF-481

Falcon 9 launched the USSF-481 mission for the U.S. Space Force from Space Launch Complex 4E at Vandenberg Space Force Base, California, at 8:01 p.m. EDT Thursday (00:01:05 UTC Friday), 17 minutes into its window, and the booster landed on its 28th flight, Spaceflight Now reported. The payload is classified. The rocket flew south, toward a polar orbit, according to Launch Library 2; GCAT logs the mission as TH-2.

Nothing else flew

No other orbital launch took place worldwide between Thursday's issue and this morning, according to Launch Library 2. Bandwagon-5, scheduled for 6:30 a.m. EDT today, was called off before Monday's countdown. "Now targeting no earlier than Tuesday, October 20 for Falcon 9's launch of the Bandwagon-5 rideshare mission from Florida due to unfavorable recovery weather conditions," SpaceX posted at 3:27 a.m. EDT.

Dragon at the station

The CRS-35 cargo Dragon, launched October 13, docked to the International Space Station (ISS) on Thursday morning and is to stay about a month (NASA).

H3 today, then a crowded week

Every orbital launch with a date between now and October 29, plus missions targeting later this month. Times are EDT, with Coordinated Universal Time (UTC) below; NET means no earlier than.

DATE / TIME	VEHICLE	PROVIDER	MISSION / PAYLOAD	SITE	ORBIT	STATUS	SOURCES
Mon Oct 19, 3:41:03 p.m. 19:41:03 UTC	H3-24	Mitsubishi Heavy Industries (MHI) for JAXA	Martian Moons eXploration (MMX), Phobos sample-return probe	LP-2, Yoshinobu complex, Tanegashima, Japan	Mars transfer	CONFIRMED	JAXA
Tue Oct 20, 6:30 a.m. 10:30 UTC	Falcon 9	SpaceX	Bandwagon-5 mid-inclination rideshare, 32 payloads, including SpaceX's second Starfall capsule demonstration. Booster B1085, 19th flight	SLC-40, Cape Canaveral, Fla.	Mid-inclination LEO	CONFIRMED	SpaceX; FAA
Tue Oct 20, 6:14 p.m. 22:14 UTC	Falcon 9	SpaceX	Starlink 15-15: 27 Starlink V2 Mini satellites. Booster B1063, 36th flight	SLC-4E, Vandenberg, Calif.	LEO	CONFIRMED	FAA; Spaceflight Now
Tue Oct 20, 11:35 p.m. 03:35 UTC Oct 21	Ceres-1S	Galactic Energy	Payload not announced	Sea launch off Rizhao, Shandong, China	Not announced	TENTATIVE	Launch Library 2
NET Wed Oct 21, 1:00 a.m. 05:00 UTC	Electron	Rocket Lab	"We're Not Hawking Around": three HawkEye 360 satellites, 98th Electron	Launch Complex 1, Mahia, New Zealand	Sun-synchronous	PRELIMINARY	Launch Library 2
Wed Oct 21, 10:57 p.m. 02:57 UTC Oct 22	Long March 8A	CASC	Payload not announced	Commercial LC-1, Hainan, China	Not announced	TENTATIVE	Launch Library 2
Fri Oct 23, 6:15 a.m. 10:15 UTC	Long March 6A	CASC	Payload not announced	LC-9A, Taiyuan, China	Not announced	TENTATIVE	Launch Library 2
Fri Oct 23, 3:25 p.m. 19:25 UTC	Long March 4B	CASC	Payload not announced	Launch Area 94, Jiuquan, China	Not announced	TENTATIVE	Launch Library 2
Sat Oct 24, 1:29 a.m. 05:29 UTC	Falcon 9	SpaceX	Starlink 15-16: 27 Starlink V2 Mini satellites	SLC-4E, Vandenberg, Calif.	LEO	TENTATIVE	FAA; Launch Library 2
NET Oct 29	Vulcan VC6L	United Launch Alliance (ULA)	Amazon Leo LV-01 broadband satellites	SLC-41, Cape Canaveral, Fla.	LEO	PRELIMINARY	Launch Library 2
NET October	GSLV Mk II	Indian Space Research Organisation (ISRO)	NVS-03 (IRNSS-1L) navigation satellite	Second Launch Pad, Sriharikota, India	GTO	PRELIMINARY	Launch Library 2
NET October	Long March 12A	CASC	Flight 2	Jiuquan, China	LEO	PRELIMINARY	Launch Library 2
NET October	SSLV	ISRO	AayulSAT-1A and others	First Launch Pad, Sriharikota, India	LEO	PRELIMINARY	Launch Library 2

CONFIRMED date and time announced by the launch provider, its customer or the FAA. **TENTATIVE** specific date in published schedules or airspace and maritime notices, not yet confirmed by the operator. **PRELIMINARY** target range, or no-earlier-than date or month only.

Abbreviations: SLC, Space Launch Complex; SLC-4E, Space Launch Complex 4 East; LP-2, Launch Pad 2; LC, Launch Complex; LEO, low Earth orbit; GTO, geostationary transfer orbit; GSLV, Geosynchronous Satellite Launch Vehicle; SSLV, Small Satellite Launch Vehicle; CASC, China Aerospace Science and Technology Corp.

Date changes

USSF-481. Listed for 7:44 p.m. Thursday; flew at 8:01 p.m. EDT, 17 minutes into the window (Spaceflight Now).

Bandwagon-5. Monday, October 19, at 6:30 a.m. on Thursday; now Tuesday, October 20, at 6:30 a.m. "due to unfavorable recovery weather conditions" (SpaceX).

H3 / MMX. Launch time unchanged; JAXA now gives a reserve period of October 21 to November 9, Japan time, against November 7 in its August plan (JAXA).

NVS-03. ISRO's October 15–20 target passed without an announced date; Launch Library 2 now lists only NET October. No reason has been given.

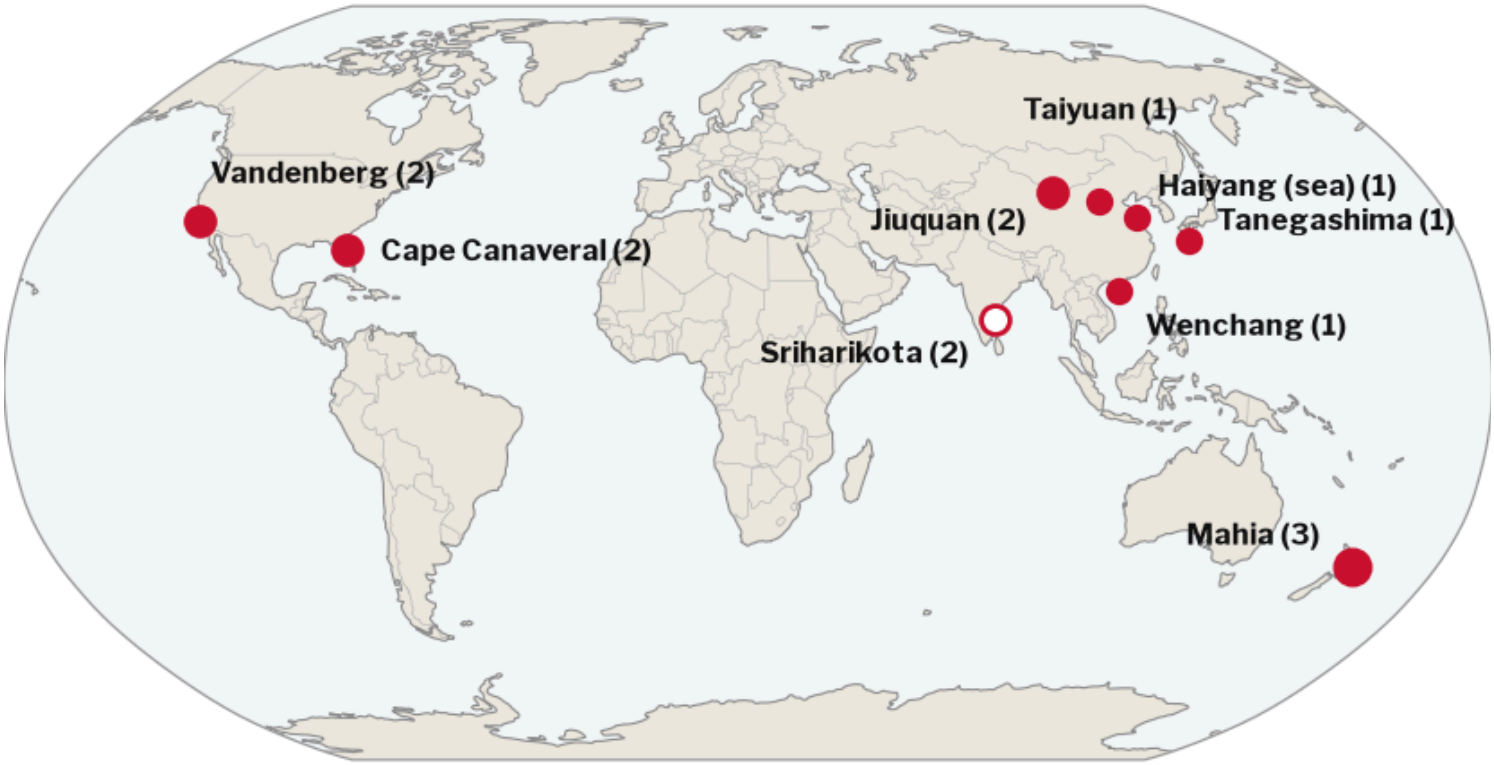
What's flying

New on the schedule. Galactic Energy's Ceres-1S sea launch and three Long March flights, an 8A from Hainan, a 6A from Taiyuan and a 4B from Jiuquan, have been added since Thursday (Launch Library 2). None of the four payloads has been announced.

Bandwagon-5. The flight also carries the second demonstration of Starfall, a SpaceX reentry capsule meant to bring experiments and in-space manufactured products back to Earth. The capsule is about 0.75 meters tall and 3.1 meters wide and can carry 1,000 kilograms, according to Launch Library 2.

QUICK HIT

Starship. Flight 15 is NET November 8 with two booster and ship catches planned, and Flight 16 is expected in December, possibly the first Starship launch from Florida's Launch Complex 39A, NASASpaceflight reported Sunday.



Launch sites with orbital launches listed through November 12; numbers count launches. Filled: dated launch (Confirmed or Tentative). Hollow: Preliminary. Vandenberg last flew on October 15. Haiyang is the port for Galactic Energy's sea launches off Rizhao.

MAP: THE LAUNCH REPORT · PAD COORDINATES FROM LAUNCH LIBRARY 2 · NATURAL EARTH

III · TRAJECTORIES & HAZARD AREAS

H3 is on the pad for MMX

The rocket rolled out Monday afternoon in Japan; its spent stages are to fall in three zones off Japan's Pacific coast.

Launch day

The H3 carrying MMX left its assembly building for the pad at about 1:30 p.m. Japan time Monday, 12:30 a.m. EDT, the Yomiuri Shimbun reported. It is the first H3 to launch a planetary probe and flies in the H3-24 configuration, with two liquid-fueled engines and four solid boosters, the most powerful version. H3 Flight 8 failed last December; flights in June and August succeeded, the paper noted. JAXA's live broadcast begins at about 2:46 p.m. EDT (JAXA).

H3 Flight 10 timeline

According to the launch plan, the rocket turns to an azimuth of 90 degrees, due east, shortly after liftoff. The upper stage fires twice and releases MMX into a Mars transfer orbit 71 minutes after liftoff. Times assume launch at 3:41:03 p.m. EDT today; events after the first second-stage cutoff shift with the launch date.

EVENT	TIME AFTER LIFTOFF	ALTITUDE
Solid rocket boosters separate (two pairs)	1:56 / 1:58	59–61 km
Payload fairing separates	3:17	131 km
Main engine cutoff (MECO)	4:52	208 km
First and second stages separate	4:59	214 km
Second-stage engine start 1 (SELI1)	5:12	224 km
Second-stage engine cutoff 1 (SECO1)	10:40	329 km
Second-stage engine start 2 (SELI2)	1:03:57	322 km
Second-stage engine cutoff 2 (SECO2)	1:10:21	497 km
MMX separates, 11.2 km/s	**1:11:12**	**592 km**

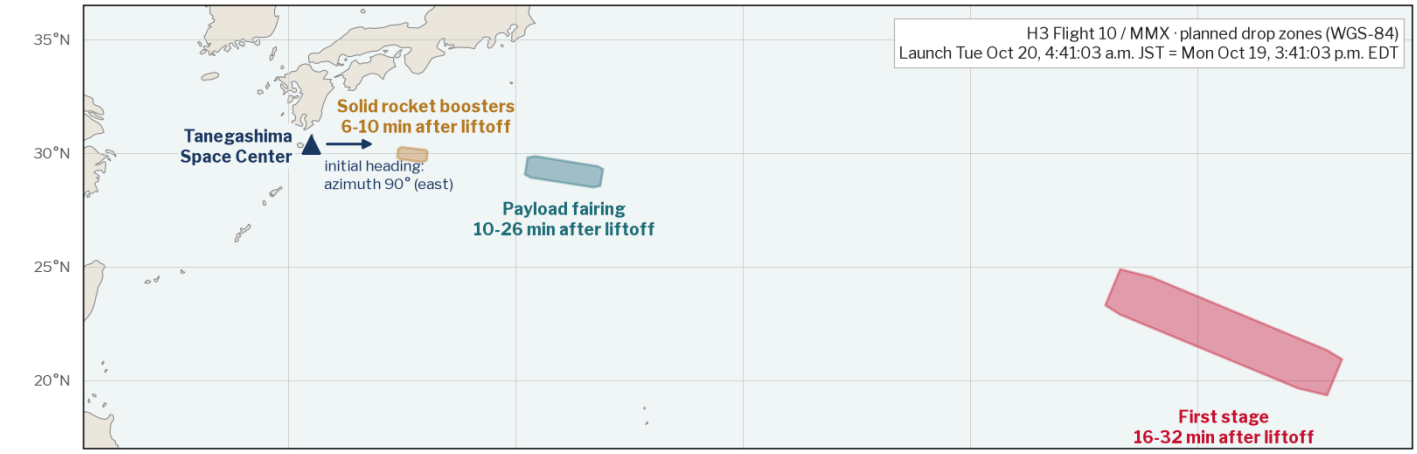


JAXA's HTV-X1 cargo spacecraft approaches the International Space Station on Oct. 29, 2025, four days after launching from Tanegashima on an H3-24, the same H3 version that will carry MMX.

PHOTO: NASA · PUBLIC DOMAIN

U.S. airspace windows

The FAA's air traffic operations plan for Monday lists Bandwagon-5 from Cape Canaveral from 6:30 to 8:13 a.m. EDT Tuesday, with a backup at the same time Wednesday. Starlink 15-15 has a window from 6:14 to 10:57 p.m. EDT Tuesday, backup from 5:59 p.m. Wednesday, and Starlink 15-16 from 1:29 to 6:12 a.m. EDT Saturday, backup from 1:14 a.m. Sunday (FAA advisory). No hazard-area coordinates have been published for these flights.



Planned drop zones for H3 Flight 10, plotted from the corner coordinates in JAXA's launch plan (World Geodetic System 1984). The solid rocket booster zone measures 62 by 128 km, the fairing zone 107 by 330 km and the first-stage zone 226 by 1,105 km. Ships and aircraft are warned to stay out of these zones through Japan's coast guard and transport ministry. The arrow shows the eastward heading the plan gives for shortly after liftoff; it is not a ground track.

MAP: THE LAUNCH REPORT · DATA FROM THE JAXA H3/MMX LAUNCH PLAN (AUG. 20, 2026) · NATURAL EARTH

IV · REENTRIES & ISS TRAFFIC

A Chinese rocket stage falls four days after launch

Four objects reentered between October 14 and 18. A Russian satellite is past its predicted time, and a Rassvet broadband satellite is next.

Confirmed reentries

CelesTrak's satellite catalog lists these objects as decayed. The Zhuque-2E stage was left in orbit by LandSpace's October 14 launch; the PSLV upper stage was predicted in Thursday's issue.

OBJECT (CATALOG NO.)	LAUNCHED	DECAYED	LAST ORBIT, KM
Starlink-30182 (57412)	Jul 20, 2023	Oct 14	208 × 222
PSLV upper stage (51659)	**Feb 14, 2022**	**Oct 15**	**130 × 145**
Starlink-5470 (54776)	Dec 17, 2022	Oct 16	136 × 143
Zhuque-2E stage (101070)	**Oct 14, 2026**	**Oct 18**	**105 × 186**

The PSLV stage came down at about 9:24 a.m. EDT on October 15, plus or minus an hour, according to Aerospace.

Predicted

Russia's Cosmos 2572 was predicted to reenter at 9:13 p.m. EDT Sunday, plus or minus 13 hours (Aerospace), but CelesTrak's latest elements, from 12:54 a.m. EDT today, still show it in a 156 by 167 km orbit. Next is Rassvet-3 17, a Russian broadband satellite launched July 19, predicted for 11:59 a.m. EDT Thursday, plus or minus 31 hours (Aerospace).



— Cosmos 2572 96.7°

One orbit of Cosmos 2572, computed from CelesTrak's orbital elements from 12:54 a.m. EDT October 19. Reentry can occur anywhere along later orbits, which shift westward each revolution. No elements were available for Rassvet-3 17.

MAP: THE LAUNCH REPORT · CELESTRAK · NATURAL EARTH

Station traffic

CRS-35 remains docked to the space-facing port of the Harmony module (NASA). The next Russian cargo ship, Progress MS-36, is listed for 12:03 a.m. Eastern Standard Time

(EST) on November 24 from Baikonur, and Northrop Grumman's Cygnus NG-25 is NET November 30 (Launch Library 2).

V · INDUSTRY & REGULATION

Rocket Lab's Neutron slips to 2027

Air pockets in a carbon-composite tank dome cost the medium-lift rocket its 2026 debut.

Neutron

Rocket Lab said Wednesday that Neutron will not fly until 2027. Inspections found voids, or air pockets, between carbon-fiber layers of a newly cured forward propellant dome for the first stage, Ars Technica reported. "They don't necessarily mean that the dome would fail in flight, but our practice is to proceed only when we believe the system has the appropriate margins," Rocket Lab founder Peter Beck said in a company video.

Engineers traced the voids to thicker composite sections added after a test tank ruptured in January. An aft dome made with a revised process has passed inspection. "We've solved the problem," Beck said. Qualification of the first stage and of the nine-engine Archimedes booster is still pending, and the rocket will then begin its launch campaign at Wallops Island, Virginia. Neutron was announced in 2021 with a first flight aimed for 2024 (Ars Technica; NASASpaceflight).

Starship plans

SpaceX removed the engines from Ship 42 for an extra cryogenic test, and Booster 22 awaits a static fire at Starbase's Pad 2 ahead of Flight 15, NET November 8, NASASpaceflight reported. Flight 16 in December would likely use Booster 23 and Ship 43. Starship has launched five times in 2026.

Comment deadlines

New Glenn at SLC-36. A virtual public meeting on raising New Glenn's cap from 12 to 50 launches a year is Thursday from 6 to 8 p.m. EDT; comments close October 30 (notice, docket FAA-2026-9841).

Part 450. Comments on five proposed changes to the commercial launch and reentry licensing rule are due November 4 (docket FAA-2026-9935).

Starbase. Scoping comments on up to 456 Starship launches a year at Starbase are due November 30, with meetings November 17 to 19 (Federal Register). No new FAA launch notices have appeared since Thursday (Federal Register).

QUICK HIT

Starshield and Ukraine. Ukrainian President Volodymyr Zelenskyy said Sunday that he believed the U.S. government was behind the speed limits on Starshield-linked vehicles and drones; Treasury Secretary Scott Bessent said no one in the Trump administration had ordered them, CNBC reported.

VI · BY THE NUMBERS

243 orbital launch attempts so far in 2026

SpaceX accounts for almost half. China needs 28 more launches to reach 100 in a year for the first time.

243	236	119	72	10
orbital launch attempts worldwide, Jan 1–Oct 15	reached orbit	by SpaceX, all successful	by China, four failures	orbital launches in October, all successful

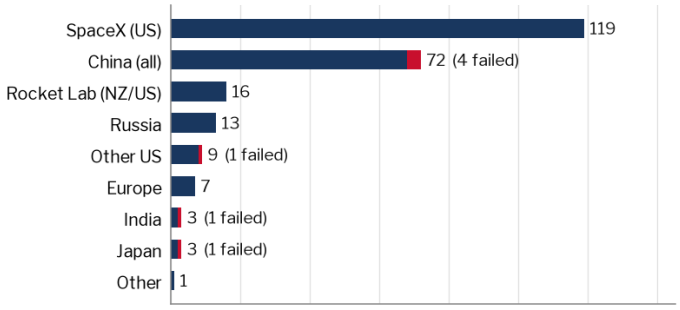
The count

Jonathan McDowell's General Catalog of Artificial Space Objects (GCAT) records 243 orbital launch attempts through USSF-481 on October 15, 236 of them successful. SpaceX has flown 119, all successful, followed by China with 72, Rocket Lab with 16 and Russia with 13. China has not launched since October 14.

Busiest launch sites

Vandenberg Space Force Base in California remains the year's busiest launch site, ahead of Florida's Cape Canaveral and Kennedy Space Center combined (GCAT, through October 15).

SITE	LAUNCHES, 2026
Vandenberg Space Force Base, California	**66**
Cape Canaveral and Kennedy Space Center, Florida	57
Jiuquan, China	27
Mahia, New Zealand (Rocket Lab)	15
Hainan commercial launch site, Wenchang, China	14
Taiyuan, China	12
Plesetsk, Russia	8



Orbital launch attempts in 2026 by provider or country. Navy: success. Red: failure or partial failure.

CHART: THE LAUNCH REPORT · GCAT (JONATHAN MCDOWELL) THROUGH OCT 15

The failures

According to GCAT, seven launches have failed this year: India's PSLV on January 12; China's Long March 3B on January 16, Ceres-2 on January 17, Tianlong-3 on April 3 and Long March 7A on August 10; Japan's Kairos on March 5; and Blue Origin's New Glenn on April 19, recorded as a partial failure.

October so far

All ten orbital launches this month have succeeded; seven were SpaceX Falcons (Launch Library 2).

DATE, EDT	ROCKET	MISSION	SITE
Oct 1, 11:10 a.m.	Falcon 9	Crew-13	Cape Canaveral
Oct 1, 2:32 p.m.	Falcon 9	Transporter-18 rideshare	Vandenberg
Oct 1, 11:54 p.m.	Falcon Heavy	NROL-97	Kennedy
Oct 6, 11:25 p.m.	Nuri	NeonSat-2 to 6	Naro, South Korea
Oct 9, 3:27 p.m.	Long March 12	SatNet LEO Group 27	Wenchang, China
Oct 10, 3:39 a.m.	Falcon 9	SDA Tranche 1 Transport Layer A	Vandenberg
Oct 12, 8:28 p.m.	Falcon 9	Starlink 15-25	Vandenberg
Oct 13, 6:33 a.m.	Falcon 9	CRS-35	Cape Canaveral
Oct 14, 7:40 a.m.	Zhuque-2E	10 satellites	Jiuquan, China
Oct 15, 8:01 p.m.	**Falcon 9**	**USSF-481**	**Vandenberg**

The week ahead

Eight orbital launches besides H3 have dates between Tuesday and Saturday in Launch Library 2. If H3 and all eight fly on schedule, October would reach 19 orbital launches by October 24, and China, idle since October 14, would add four. Three of the eight are SpaceX Falcon 9s, which would bring the company's 2026 total to 122.

Takeaways

1. USSF-481 was the only orbital launch worldwide since Thursday.
2. H3 launches MMX toward Mars at 3:41 p.m. EDT today; JAXA's reserve period runs to November 9.
3. Bandwagon-5 slipped to 6:30 a.m. EDT Tuesday because of recovery-zone weather.
4. Rocket Lab moved Neutron's first flight to 2027 after finding voids in a tank dome.
5. Cosmos 2572 is still in orbit past its predicted reentry time.

KEY DATES AHEAD (EDT)

Mon Oct 19	H3 / MMX, 3:41 p.m.
Tue Oct 20	Bandwagon-5, 6:30 a.m.; Starlink 15-15, 6:14 p.m.; Ceres-1S, 11:35 p.m.
Wed Oct 21	Electron for HawkEye 360, NET 1 a.m.; Long March 8A, 10:57 p.m.
Thu Oct 22	Rassvet-3 17 reentry predicted, about noon; New Glenn cadence meeting, 6–8 p.m.
Fri Oct 23	Long March 6A, 6:15 a.m.; Long March 4B, 3:25 p.m.
Sat Oct 24	Starlink 15-16, 1:29 a.m.
NET Oct 29	Vulcan returns with Amazon Leo
NET Nov 8	Starship Flight 15

SOURCES

Schedules: Launch Library 2 (The Space Devs), Spaceflight Now, NASASpaceflight, FAA operations plan advisory. Agencies: NASA, JAXA, FAA, Federal Register. Tracking: CelesTrak, The Aerospace Corporation, GCAT (Jonathan McDowell). Reporting: Ars Technica, The Yomiuri Shimbun, CNBC. X posts: @SpaceX. Photos: NASA (public domain). Maps: Natural Earth (public domain).

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